

NOWCAST

AI-Driven Hyperlocal Severe-Weather Nowcasting

Smart India Hackathon 2024

ML + Inference Track

Tech Stack & Project Overview

Pilot Region: Uttarakhand & Delhi NCR | Grid: 0.1° (~11 km) | Hazards: Thunderstorm · Cloudburst
· Flash Flood | Lead Time: 1–6 hours

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1. Problem Statement

1.1 The Challenge

India's Himalayan foothills and the Delhi NCR region are among the world's most vulnerable areas for **extreme weather events**: thunderstorms, cloudbursts (≥ 100 mm/h in 1 hour), and flash floods. These events develop over **sub-hourly timescales**, often missing standard synoptic warning windows of 12–24 hours.

Conventional Numerical Weather Prediction (NWP) models such as WRF or the operational IMD suite run at coarse spatial resolution (≥ 25 km) and are computationally expensive to execute quickly enough for **very-short-range nowcasting** (0–6 hour forecasts). By the time a NWP alert is issued and disseminated, the event may already be in progress.

1.2 Impact

- **Loss of life & property:** Flash floods in Uttarakhand (2013, 2021, 2023) caused thousands of casualties and billions of rupees in infrastructure damage.
- **Early warning gaps:** IMD issues district-level warnings but lacks the grid-level spatial specificity needed for targeted evacuations.
- **Limited nowcast lead time:** Current operational systems provide ≤ 1 h advance notice for cloudbursts—insufficient for effective response.
- **Data heterogeneity:** Merging reanalysis (IMDAA/MERA), satellite (INSAT-3D/3DR), and precipitation (IMERG) data at a common grid is a significant engineering challenge.

2. Our Solution

We built **Nowcast**: a fully data-driven, deep-learning pipeline that fuses multi-source atmospheric data into hyperlocal, probabilistic hazard maps updated every hour. The system predicts the *probability of occurrence* of thunderstorm, cloudburst, and flash flood events at each 0.1° grid cell, for lead times of 1 through 6 hours ahead.

2.1 Core Idea

Feed 12 hours of multi-source atmospheric & terrain features into a spatiotemporal deep neural network → simultaneously predict 3 hazard probability maps $\times$ 6 lead times = 18 output grids per inference run.

2.2 What Makes It Different

Capability	Traditional NWP	Our System
Spatial resolution	25–100 km	~11 km (0.1°)
Time to forecast	1–3 hours compute	< 1 minute inference
Lead times	12–72 hours	1–6 hours (nowcast sweet spot)
Terrain coupling	Parametrised	Explicit DEM routing (D8 algorithm)
Uncertainty	Ensemble-only	Temperature scaling calibration
Satellite fusion	Partial / offline	Real-time INSAT-3D/3DR CTT + WV
Output type	Deterministic fields	Per-cell probability maps

3. Data Sources

The system ingests four independent data streams, harmonised to a common **0.1° lat-lon grid** at **1-hourly time steps** via an Xarray/Zarr datacube.

Source	Type	Variables	Resolution	Role
IMDAA	Reanalysis	T, U, V, Z, RH (7 levels), T2m, U10, V10, PRMSL, TCWV, CAPE, CIN	0.1°, 3h	Primary atmospheric state
MERA	Reanalysis	Supplementary surface/upper-air fields	0.1°, 3h	Gap-fill / cross-check
INSAT-3D/3DR	Satellite	Cloud-Top Temperature (TIR-1 BT), Water Vapour BT	0.1°, 30 min	Real-time convective proxy
GPM IMERG	Satellite Precip	Precipitation rate (mm/h)	0.1°, 30 min	Label source + precip feature
Copernicus GLO-30 DEM	Slope/Terrain	Elevation, slope, flow accumulation, flow direction, HAND	30m, 1h5s	Flash-flood routing context

4. Feature Engineering

Raw datacube variables are transformed into **20 physically-motivated channels** that the neural network ingests as a (C=20, T=12, H=38, W=49) tensor per sample.

4.1 Feature Channels (in model order)

#	Channel	Physics / Description	Source
1	t2m	2 m air temperature (K) — boundary-layer instability	IMDAA
2	prmsl	Mean sea-level pressure (Pa) — synoptic-scale forcing	IMDAA
3	tcwv	Total column water vapour (kg m^{-2}) — moisture supply	IMDAA
4	cape	CAPE (J kg^{-1}) — convective available potential energy	IMDAA/MetPy
5	cin	CIN (J kg^{-1}) — convective inhibition (lid strength)	IMDAA/MetPy
6	iwv	Integrated Water Vapour — trapz over 7 pressure levels	IMDAA
7	iwv_tendency_1h	1 h IWV tendency — rapid moistening signal	Derived
8	lifted_index	Lifted Index — parcel vs environment temperature at 500 hPa	MetPy
9	conv_850	850 hPa wind convergence — low-level moisture flux	MetPy
10	shear_0_6km	0–6 km bulk wind shear (m s^{-1}) — storm organisation	MetPy
11	shear_0_1km	0–1 km low-level shear — tornado/waterspout risk	MetPy
12	u10	10 m zonal wind	IMDAA
13	v10	10 m meridional wind	IMDAA
14	ctt	Cloud-Top Temperature (K) — convective depth proxy	INSAT-3D
15	ctt_drop_rate_1h	1 h CTT decrease rate — rapidly deepening convection	Derived
16	ww_bt	Water-vapour channel brightness temp — upper-troposphere moisture	INSAT-3D
17	elevation	Terrain elevation (m) — orographic lifting trigger	GLO-30 DEM
18	slope	Terrain slope (°) — runoff amplification	GLO-30 DEM
19	flow_accumulation	D8 upstream cell count ($\log_{10}$ normalised)	pysheds
20	hand	Height Above Nearest Drainage (m) — flood exposure	pysheds

4.2 Label Construction

Target labels are **peak-normalised Gaussian-smoothed occurrence probability maps**. For each hazard, a binary threshold mask is applied to the relevant diagnostic variable (precipitation rate, upstream accumulation, or CTT), then smoothed with $\sigma=1$ grid cell and peak-normalised so an isolated event cell reaches exactly 1.0.

Hazard	Primary Threshold	Secondary Condition
Thunderstorm	Precip ≥ 10 mm h ⁻¹	CTT ≤ 241 K
Cloudburst	Precip ≥ 50 mm h ⁻¹ OR 60 mm in 2 h	—
Flash Flood	Upstream routed accumulation ≥ 80 mm	D8 drainage catchment routing

5. Model Architecture

5.1 MultiTaskNowcastNet

The core deep model is a **multi-task network** with one shared spatiotemporal backbone and one hazard-specific prediction head per target. This design allows all hazard predictors to share atmospheric-pattern representations while maintaining task-specific outputs.

Input: (B, C=20, T=12, H=38, W=49) → Backbone → Feature Maps → 3 Heads → Output: (B, H_hazards=3, L_leads=6, H=38, W=49)

5.2 Backbone — SpatiotemporalBackbone

- **ConvLSTM blocks:** Capture temporal evolution of atmospheric fields. Each ConvLSTM cell processes the full (H, W) feature map with convolutional gates, preserving spatial structure across the 12-step input sequence.
- **Transformer encoder (optional):** A spatial self-attention layer applied to the ConvLSTM output features, allowing long-range spatial dependencies to be modelled (e.g. upstream moisture tracking across the Himalayan foothills).
- **Residual skip connections:** Stabilise gradient flow across the deep network.
- **Output:** A compressed feature map (B, F, H, W) summarising spatio-temporal atmospheric state.

5.3 Hazard Heads

Head	Type	Extra Input	Output
Thunderstorm	HazardHead (Conv2D + upsampler)		(B, 6, 38, 49) logits
Cloudburst	HazardHead (Conv2D + upsampler)		(B, 6, 38, 49) logits
Flash Flood	FlashFloodHead (Conv2D + ternary ops)	Training inputs: flow_accumulation, flow_direction, Hgms	(B, 6, 38, 49) logits

5.4 Loss Function — MultiTaskLoss

Training uses a composite loss that addresses the extreme class imbalance (positive cells ≈ 1 in 10,000 cell-hours):

- **Binary Cross-Entropy (BCE)** with pos_weight=20 — lifts gradient contribution of rare positive cells.
- **Dice Loss** (weight=0.5) — region-overlap term; robust to imbalance, trains the model to cover the event footprint.
- **Focal Loss** ($\alpha=0.25$, $\gamma=2.0$, weight=1.0) — Lin et al. (2017); down-weights easy-correct negatives, focuses learning on hard examples.
- All three losses train against the **soft (probabilistic) label**; CSI/POD/FAR binarise at threshold 0.5 for evaluation.

5.5 LightGBM Baseline

As a strong non-deep-learning reference, we train a **LightGBM gradient-boosted tree** on pixel-level feature vectors (flattening the spatial grid). This baseline:

- Uses the same 20-channel feature set per grid cell per time step.
- Provides interpretable feature importances (SHAP values) for model debugging.
- Sets a CSI/POD/FAR benchmark that the deep model must exceed to justify its complexity.

- Runs end-to-end significantly faster than the ConvLSTM, making it a useful rapid iteration and ablation tool.

5.6 Uncertainty Quantification

- **Temperature scaling** (Guo et al., 2017): a single learnable parameter T re-calibrates logit magnitudes post-training so that predicted probabilities match observed event frequencies.
- Reliability diagrams and ECE (Expected Calibration Error) track calibration quality.
- Calibration is fitted on the validation split using hard-binarised labels at the occurrence threshold.

6. Technology Stack

6.1 Backend — Machine Learning & Data

Library / Tool	Version	Purpose
Python	≥ 3.10	Core language
PyTorch	≥ 2.2	Deep learning framework (ConvLSTM, Transformer, loss functions)
PyTorch Lightning	≥ 2.2	Training orchestration, checkpointing, multi-GPU support
TorchMetrics	≥ 1.3	CSI, POD, FAR, PR-AUC metrics with GPU support
LightGBM	≥ 4.3	Gradient-boosted baseline model
scikit-learn	≥ 1.4	Preprocessing, calibration utilities, train/test split guards
NumPy	≥ 1.26	Array operations, D8 routing, label construction
SciPy	≥ 1.11	Gaussian smoothing (label construction), statistical utilities
Xarray	≥ 2024.3	Labelled N-D array datacube (time × level × lat × lon)
Dask	≥ 2024.5	Out-of-core / parallel Xarray computation
Zarr	≥ 2.17, < 3	Chunked on-disk datacube storage format
NetCDF4 / h5py / h5netcdf	≥ 1.6 / ≥ 3.10 / ≥ 1.3	IMDAA, MERA, INSAT HDF5 / NetCDF file I/O
MetPy	≥ 1.6	Lifted Index, IWV, convergence, bulk wind shear calculations
MLflow	≥ 2.12	Experiment tracking, run comparison, artifact logging
Prefect	≥ 2.19	Ingest pipeline orchestration, task scheduling
Pydantic	≥ 2.6	Config validation and data contract enforcement

6.2 Geospatial Stack

Library	Version	Purpose
rioxarray	≥ 0.15	GeoTIFF / COG read/write, CRS-aware raster operations

Library	Version	Purpose
rasterio	≥ 1.3.9	Low-level raster I/O, reprojection, resampling
pyproj	≥ 3.6	CRS transformations (EPSG:4326 target grid)
Shapely	≥ 2.0	Geometry operations, bounding-box clipping
pysheds	≥ 0.4	D8 flow-direction routing, flow accumulation, HAND computation

6.3 Data Fetching

Library	Purpose
cdsapi ≥ 0.7	Download MERA / ERA5 from Copernicus Climate Data Store
earthaccess ≥ 0.9	NASA Earthdata authentication + GPM IMERG HDF5 download
pooch ≥ 1.8	Verified URL download with hash checking (Copernicus GLO-30 DEM tiles)
requests ≥ 2.31	HTTP utility for NCMRWF IMDAA / INSAT data endpoints

6.4 Frontend — Web Dashboard (Nowcast Console)

An interactive operator console visualises nowcast output as animated probability heat-maps overlaid on a geographic base map.

Technology	Version	Purpose
React	^18.3.1	Component-based UI framework
Vite	^5.4.0	@vitejs/plugin-react — fast dev server + optimised bundle
MapLibre GL JS	^4.7.1	WebGL-accelerated interactive base map (OSM / satellite tiles)
deck.gl	^9.1.0	GPU-rendered probability overlay layers (HeatmapLayer, GridLayer)
@deck.gl/mapbox	^9.1.0	deck.gl ↔ MapLibre GL interop
@deck.gl/layers	^9.1.0	Scatterplot, heatmap and grid layer primitives

6.5 Development & Quality

Tool	Purpose
pytest ≥ 8.1 + pytest-cov ≥ 5.0	149-test synthetic-data test suite; shape/schema/loss/metric correctness
Ruff ≥ 0.4	Linting (E, F, I, UP, B, W rules) + import sorting; zero-warning policy
Git / GitHub	Version control, slice-based branch workflow
MLflow	Hyperparameter logging, run comparison, model registry
Setuptools ≥ 68	Editable package install (pip install -e .)

7. System Architecture

The system is structured into four loosely-coupled **slices** ($A \rightarrow B \rightarrow C \rightarrow D$), each owning a specific responsibility and communicating through schema-validated data contracts.

Slice	Module	Responsibility	Key Files
A — Data	nowcast/ingest/ nowcast/dem/ nowcast/pipelines/	Fetch raw data, parse IMDAA/MERA/IMERG/INSAT for files, normalize DEM, coding, assemble	
B — Features	nowcast/features/	Compute all 20 physics-derived channels, build persistence, eddy, lateral, psychrosol	
C — Model	nowcast/models/ nowcast/data/ nowcast/training/ nowcast/uncertainty/	MultiTaskNowcastNet (backbone + heads), Lightning base, lightning, training, calibration	
D — Serving	frontend/	React + deck.gl operator console; fetches inference, complex and renders configured probability over	

7.1 Data Flow

- **Ingest (Prefect pipeline):** Raw files → IMDAA/MERA/IMERG/INSAT parsers → tolerance-bounded time alignment → regrid to 0.1° → merge into Zarr datacube.
- **DEM pre-processing:** GLO-30 GeoTIFF → pysheds routing → elevation, slope, flow_accumulation, flow_direction, HAND static layers.
- **Feature assembly:** Datacube slice (12 h window) → MetPy/NumPy transforms → normalised (C, T, H, W) tensor; terrain planes normalised separately ($\log_{10}$ for flow_accumulation, z-score otherwise).
- **Training:** PyTorch Lightning Trainer → event-balanced WeightedRandomSampler (35% event windows) → MultiTaskLoss → Adam → MLflow logging → best checkpoint.
- **Calibration:** Held-out validation set → Temperature Scaling → calibrated probability output.
- **Inference → Frontend:** Calibrated model → JSON/GeoJSON probability grid → deck.gl HeatmapLayer overlay on MapLibre base map.

7.2 Chronological Data Split

Split	Date Range	Purpose	Note
Train	2018-04-01 → 2018-07-31	Model parameter learning	Includes July (most event-dense month); no data
Validation	2018-08-01 → 2018-08-25	Hyperparameter selection, early stopping	Disjoint; fail-loud guard if overlapping
Test	2018-08-26 → 2018-09-30	Final held-out evaluation (quoted in docs)	Disjoint; never used during training

8. Evaluation Metrics

Metric	Formula	Interpretation
CSI (Critical Success Index)	$TP / (TP + FP + FN)$	Hits over all events; the standard verification score for rare-event forecasting

Metric	Formula	Interpretation
POD (Probability of Detection)	$TP / (TP + FN)$	Fraction of observed events that were forecast
FAR (False Alarm Ratio)	$FP / (TP + FP)$	Fraction of forecast events that were false alarms
PR-AUC	Area under Precision–Recall curve	Threshold-free; best summary for severely imbalanced classes
ECE (Expected Calibration Error)	Averaged mean confidence – accuracy	Probability calibration quality (lower = better)

9. Project Structure

Project - SIH/	
nowcast/	← Python package (ML + inference)
config.py	← Single source of truth: grid, thresholds, paths
schema.py	← Datacube & tensor data contract; validators
ingest/	← IMDAA, MERA, IMERG, INSAT parsers; datacube builder
dem/	← GLO-30 DEM download & pysheds routing
features/	← 20-channel assembly; label construction; MetPy transforms
models/	← MultiTaskNowcastNet, backbone, heads, transformer
data/	← PyTorch Lightning DataModule; NowcastDataset; transforms
training/	← LightningModule; MultiTaskLoss; metrics; train.py
baseline/	← LightGBM train/eval; pixel-dataset builder
uncertainty/	← TemperatureScaler; reliability diagram; ECE
pipelines/	← Prefect ingest_flow; orchestration DAG
common/	← Shared helper utilities
frontend/	← React + deck.gl operator console
src/	← Components, map layers, API hooks
static-mockup/	← Static HTML prototype / design reference
DataSet/	← Raw, interim, processed data (gitignored)
raw/	← imdaa/, mera/, imerg/, insat3d/, dem/
interim/	← Per-source intermediate Zarr stores
processed/	← datacube.zarr, dem_routing.zarr, norm_stats.json
tests/	← 149-test pytest suite (synthetic data only)
docs/	← Review docs, this PDF
requirements.txt	← All Python dependencies
pyproject.toml	← Build system, Ruff, pytest config

10. Current Status & Roadmap

10.1 Completed (Reviewed & Verified)

- Schema-driven datacube contract with validation (schema.py)
- IMDAA/MERA/IMERG/INSAT ingest parsers with time-alignment and accumulation-reset handling
- DEM download and D8 routing pipeline (elevation, slope, flow_acc, flow_dir, HAND)
- All 20 feature channels including MetPy-derived thermodynamic/kinematic indices
- Peak-normalised label construction for all 3 hazards × 6 lead times
- MultiTaskNowcastNet (ConvLSTM backbone + 3 hazard heads + FlashFloodHead terrain branch)
- MultiTaskLoss (BCE + Dice + Focal) with event-balanced sampler
- LightGBM baseline with chronological split guards
- Temperature scaling calibration; ECE; reliability diagrams
- CSI/POD/FAR/PR-AUC metrics (NaN-safe on degenerate splits)
- 149-test synthetic test suite; Ruff linting; MLflow tracking
- React + deck.gl frontend mockup (probability overlay console)

10.2 In Progress / Pending

Item	Priority	Description
F19 — Temporal leakage fix	BLOCKING	NowcastDataset must exclude label windows that straddle a split boundary (already fixed)
Real data ingestion	HIGH	IMDAA/MERA APCP convention verification; INSAT L1C HDF5 LUT; IMERG 30 min → 15 min
Flow accumulation normalisation	MEDIUM	Shared backbone flow_accumulation channel needs log ₁₀ (not plain z-score)
Norm-stats cache versioning (F20)	LOW	Key norm_stats.json by hash of (FEATURE_CHANNELS, TRAIN_DATE_RANGE, datacube)
0.1° routing resolution (F9)	LOW / Deferred	Pysheds routing on native DEM resolution before regrid for accurate catchments
Helper deduplication (F12)	LOW / Deferred	Promote duplicated helpers to nowcast/common/
Validity mask (F13)	LOW / Deferred	Thread missing-data mask through assemble → labels → loss

10.3 Future Work

- **Real-time inference API:** FastAPI endpoint serving nowcast grids every hour; integration with IMD alert dissemination systems.
- **Extended data window:** Incorporate 2019–2023 IMDAA/MERA/IMERG for multi-year training, covering more extreme events.
- **Higher resolution:** Downscale to 0.05° (~5 km) using terrain-conditioned super-resolution.
- **Ensemble output:** Monte Carlo dropout or deep ensembles for richer uncertainty representation.
- **CAPE nowcasting:** Replace MetPy-estimated CAPE with direct IMDAA CAPE field for better accuracy at all 7 pressure levels.
- **Operational integration:** Pilot with NDMA / SDMA for real-time alert generation during the pre-monsoon and monsoon seasons.

11. References & Acknowledgements

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